including genetic diseases and cancer. Some banned chemical warfare agents (e.g. mustard gas) also are mutagens.

A number of chemicals (subclass of mutagens) induce mutations that result in tumorous or cancerous growth. These chemical agents are called chemical carcinogens. Directly testing the chemicals for their ability to cause tumors in animals is time consuming and expensive. However, the fact that most chemical carcinogens are mutagens led Bruce Ames to develop a simple, inexpensive, indirect assay for mutagens. In general Ames test is an indicator of whether the chemical is a mutagen. The Ames test assays the ability of chemicals to revert mutant strains of the bacterium *Salmonella typhimurium* to wild type. The mutant strain of *S.typhimurium* is auxotrophic to histidine (his^-), that is it requires histidine for its growth and cannot grow in the absence of histidine. The mutant strain is grown in a histidine deficient medium containing the chemical to be tested. A control plate is also set up which does not contain the chemical. After incubation the control plates may have few colonies resulting from spontaneous reversion of the his- strain. Compared to the control plates if there are increased number of colonies on test plate, it indicates that the chemical has reverted the mutant strain back to wild type. This chemical is likely to be a carcinogen. Figure 12.14 shows steps in Ames test.

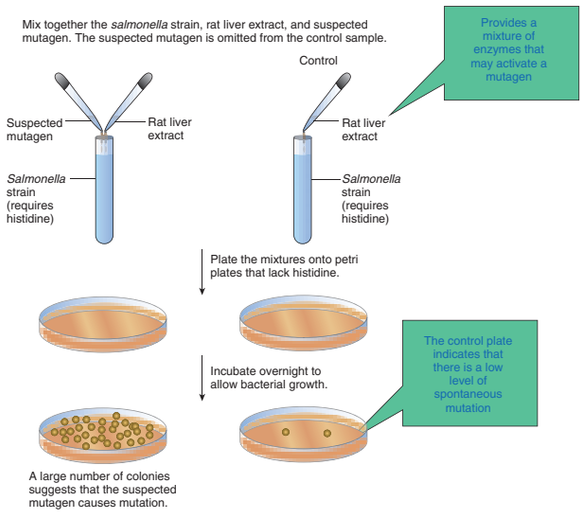


Figure 12.14: Steps in Ames test

Infobits

DNA Repair

Both prokaryotes and eukaryotes have a number of repair systems that deal with different kinds of DNA damage. All the systems use enzymes to make correction. Without this repair systems lesions would accumulate and be lethal to the cell or organism. Not all lesions are repaired, and mutations do appear, but at low frequencies. At high doses of mutagens, repair systems are unable to correct all of the damage, and cell death may result. We can group repair systems into different categories on the basis of the way they operate. Some systems correct damaged areas by reversing the damage. This type of repair is called direct correction or direct reversal. Other systems excise the damaged areas and then repair the gap by new DNA synthesis. Some of the DNA repair systems are

- Mismatch repair by DNA polymerase proofreading
- Repair of UV induced pyrimidine dimers- Photo reactivation or Light repair
- Base excision repair
- Nucleotide excision repair

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